ETALEITY DETECTED FROM THE COMPOSITION AT A UNIBRANCH INTEGRAL BASE WITH A SPECIALIZATION FROM THE GENERIC POINT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10529-10540 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10542-10553, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Unramified permanence under composition (Morphisms), the etale-at-a-point criterion for unramified dominant maps over unibranch bases (same chapter), and etale descent of the etale property (Descent); cited inputs with the four-step argument written in the retained span.

Source reference. Original lemma 02KM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10553–10553. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GS8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10466–10475. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10553–10553. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GG: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10553–10553. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of schemes. Let $x \in X$ with image $y \in Y$. Assume*

(1) *Y is integral and geometrically unibranch at y,*
(2) *$g \circ f$ is étale at x,*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(3) *there is a specialization $x' \rightsquigarrow x$ such that $f(x')$ is the generic point of Y.*

Then f is étale at x and g is étale at y.

Proof. After replacing X by an open neighbourhood of x we may assume $g \circ f$ is étale. Then we find f is unramified by Morphisms, Lemmas 02GG and 02GK. Hence f is étale at x by Lemma 0GS8. Then by étale descent we see that g is étale at y, see for example Descent, Lemma 02KM. □